From Möbius Time to Circular/Spiral Time: Event–Coded Dynamics under Decimated Transfer–Matrix Constraints

VR & AI

August 24, 2025

Abstract

We propose a simple, self–consistent model of *temporal envelopes* that change topology from a Möbius phase (orientation–reversing macrocycle) to a circular or spiral phase (standing vs. running) under event injection and relaxation. The dynamics are constrained by an l–step decimation identity derived from Cayley–Hamilton on the unit–cell transfer monodromy M^l: $x_{k+2l} - V_l x_{k+l} + Q^l x_k = 0$ with $V_l = \mathrm{tr}(M^l)$ and $Q^l = \det(M^l)$. Events are *thresholded increments of integrated power/flux* measured at a fixed cadence; decoded integers satisfy a *Chebyshev–Lucas checksum.* A *half–twist macrocycle* (π holonomy) realizes Möbius time; a *quench* controlled by a dimensionless parameter Λ (tick vs. normalization time) transitions the system to circular (standing) or spiral (running) regimes. We give clear operators, invariants, and selection rules (hard/soft decoding), prove log–growth bounds for the spiral pitch through the decimated roots, and describe reproducible protocols in digital, microwave, and acoustic chains. The framework is minimal: topology enters via boundary conditions/holonomy, dynamics via transfer matrices, and robustness via a finite–state trellis decoder.

Contents

1 Motivation and contributions

Temporal envelopes. We model "time shape" not as a new metric, but as a *layout* of stroboscopic samples on a 2D plane: a circle (standing), a logarithmic spiral (running), or a figure–8 macrocycle with a half–twist (Möbius–like). The layout is visualization; the physics lives in the *transfer* of samples and the *events* we inject/measure.

Contributions.

- A *minimal dynamical core*: an l–step decimation constraint from the unit–cell transfer matrix M: $x_{k+2l} - V_l x_{k+l} + Q^l x_k = 0,\ V_l = \mathrm{tr}(M^l),\ Q^l = \det(M^l)$.
- A *Möbius macrocycle*: an orientation–reversing holonomy (π) constructed from blocks; decimation invariants survive the twist.
- An *eventization/decoding layer*: events are thresholded increments of integrated power/flux; decoded digits obey a Chebyshev–Lucas checksum; hard vs. soft trellis decoders with selection/tuning rules.
- A *mode switch* governed by Λ (tick vs. normalization); circular (standing) vs. spiral (running) phases.

- *Pitch law*: spiral support radius grows like $R(T) = \Theta(\log_r T)$ with r the large root of $z^2 - V_l z + Q^l = 0$; the helix pitch in the plane follows from $R(T)$ and the sampling angle.
- *Protocols*: platform–agnostic steps to realize, observe, and test the topology switch and the checksum.

2 Core constraint from transfer matrices

2.1 Second–order chain and l–decimation

A one–dimensional chain with nearest–neighbor coupling reduces after eliminating a conjugate variable to

$$x_{k+1} = Px_k - Qx_{k-1}, \qquad P, Q \in \mathbb{C},\ Q \neq 0, \tag{1}$$

equivalently

$$\begin{pmatrix} x_{k+1} \\ x_k \end{pmatrix} = M \begin{pmatrix} x_k \\ x_{k-1} \end{pmatrix}, \qquad M = \begin{pmatrix} P & -Q \\ 1 & 0 \end{pmatrix}, \quad \mathrm{tr} M = P,\ \det M = Q. \tag{2}$$

Cayley–Hamilton on M^l yields the l–decimated tri–site identity

$$x_{k+2l} - V_l x_{k+l} + Q^l x_k = 0, \qquad V_l := \mathrm{tr}(M^l), \quad Q^l := \det(M^l) = (\det M)^l. \tag{3}$$

2.2 Time topology as boundary condition: PBC vs. APBC

On a ring (finite k mod L), *periodic* boundary conditions (PBC) enforce monodromy $+1$. *Anti–periodic* boundary conditions (APBC, π–flux) enforce a *half–twist* (holonomy -1): after one round–trip, the local phase flips. APBC instantiate a Möbius–like time on a discrete ring. The decimated constraint (**??**) is local and survives either BC.

3 Layouts in the plane: circle, spiral, and figure–8

Fix a stride l and an angle step $\Delta\theta = 2\pi/l$. Define a *complex base* $B = b\, e^{i\Delta\theta}$ with $b > 1$. We lay out sites k in the plane by $z_k = B^k$ (up to a global scale), so consecutive samples rotate by $\Delta\theta$ and expand by b.

- **Circle (standing):** choose $b = 1$; samples lie on a circle (no radial drift).
- **Logarithmic spiral (running):** choose $b > 1$; samples lie on a logarithmic spiral (constant angle step, geometric radial growth).
- **Figure–8 macrocycle:** two blocks with net rotation π and unit radial gain realize an orientation–reversing loop; see Sec. **??**.

The geometry is a visualization; dynamics are enforced by (**??**).

4 Events, decoding, and invariants

4.1 Eventization (operational)

At cadence Δt and site t, let $E_t[n]$ be integrated power/flux in window n and fix a quantum $\varepsilon > 0$. Define event increments

$$\delta N_t[n] = \left\lfloor \frac{E_t[n] - E_t[n-1]}{\varepsilon} \right\rfloor \in \mathbb{Z}_{\geq 0}, \qquad T_s := \sum_n \delta N_s[n]. \tag{4}$$

4.2 Chebyshev–Lucas checksum

Let r be a root of $z^2 - V_l z + Q^l = 0$ with $|r| \geq 1$ and define $I(d) = \sum_k d_k r^k$ for finite integer sequences d. Tri–site moves $(d_{t-1}, d_t, d_{t+1}) \mapsto (d_{t-1} + Q^l,\ d_t - V_l,\ d_{t+1} + 1)$ preserve $I(d)$, hence

$$\sum_k d_k r^k \ = \ \sum_s T_s\, r^s. \tag{5}$$

With anchor events only ($s = 0$) we get an integer checksum $\sum_k d_k r^k = T$. On a symmetric window $[-R, R]$ with $d_{-t} = d_t$,

$$\sum_{k=-R}^{R} d_k r^k = d_0 + \sum_{j=1}^{R} d_j\, D_j(V_l, Q^l), \quad D_{j+1} = V_l D_j - Q^l D_{j-1},\ D_0 = 2,\ D_1 = V_l. \tag{6}$$

4.3 Decoders: selection and tuning

We observe $m_t = d_t + \eta_t$ on a window $[-R, R]$ and recover integers $d_t \in \mathcal{D}$. Let $e_t = d_{t+1} - V_l d_t + Q^l d_{t-1}$. We use:

$$\textbf{Soft:} \quad \hat{d} \in \arg\min_{d \in \mathcal{D}^{2R+1}} \sum_{t=-R}^{R} w_t |d_t - m_t|^2 + \lambda \sum_{t=-R}^{R-1} |e_t|^2, \tag{7}$$

$$\textbf{Hard:} \quad \hat{d} \in \arg\min_{d \in \mathcal{D}^{2R+1}} \sum_{t=-R}^{R} w_t |d_t - m_t|^2 \ \text{ s.t. } e_t = 0\ \forall t. \tag{8}$$

Selection: use hard when the window sits in a periodic/twisted segment (nontrivial kernel present) and model mismatch $\ll$ noise; use soft otherwise. *Tuning λ:* balance residuals, use $\lambda \approx \sigma^2/\sigma^2_{\text{mod}}$, or L–curve.

Both problems admit a finite–state trellis (states $(d_{t-1}, d_t) \in \mathcal{D}^2$), solved by dynamic programming in $O(R\,|\mathcal{D}|^2)$ time. Design rules: $A = \max\{A_{\min}(R), \lfloor\sqrt{|V_l|}\rfloor\}$ with $A_{\min}(R) = \lceil T_{\max}/(1 + \sum_{j=1}^{R} D_j)\rceil$, and $R = \lceil \log_r(1 + \frac{(r-1)T_{\max}}{A})\rceil - 1 + m$ with $m \in \{2, 3\}$.

5 Half–twist macrocycle (Möbius time)

Partition one macrocycle into J consecutive blocks $j = 1, \ldots, J$. Block j advances by $m_j \in \mathbb{N}$ steps of $B_j = b_j e^{i2\pi/l_j}$ and applies a phase gate θ_j and a positive block scale S_j. The macrocycle transform is

$$T \ := \ \Big(\prod_{j=1}^{J} S_j b_j^{m_j}\Big)\ \exp\Big(i\,2\pi \sum_{j=1}^{J} \tfrac{m_j}{l_j}\Big)\ e^{\,i\sum_j \theta_j}. \tag{9}$$

Möbius (half–twist) condition. Choose (m_j, S_j, θ_j) so that

$$\prod_j S_j b_j^{m_j} = 1, \qquad \sum_{j=1}^{J} \frac{m_j}{l_j} + \frac{1}{2\pi}\sum_j \theta_j \equiv \frac{1}{2} \pmod 1. \tag{10}$$

Then $T = -1$: the macrocycle returns to the same point with reversed local phase (APBC). The decimated constraint (**??**) is local and unaffected.

6 Mode switch: circular vs. spiral

Let τ_{tick} be the event cadence and τ_{norm} the normalization time per step (trellis update/carry relaxation). Define $\Lambda := \tau_{\text{norm}}/\tau_{\text{tick}}$.

- **Standing (circular).** For small Λ (fast relaxation), decoded digits equilibrate within each macrocycle, producing $b \approx 1$ (no radial drift).
- **Running (spiral).** For larger Λ, residual load carries across macrocycles; effective $b > 1$ and the layout runs along a logarithmic spiral.

The *Chebyshev–Lucas invariant* (**??**) holds in either mode (after decoding).

7 Spiral pitch and growth laws

Let $r > 1$ be the large root of $z^2 - V_l z + Q^l = 0$ and let T be the total anchor events accumulated. On a window, the occupied radius obeys

$$\left\lceil \log_r\Big(1 + \frac{(r-1)T}{2A}\Big)\right\rceil - 1 \;\le\; R(T) \;\le\; \left\lceil \log_r\Big(1 + \frac{(r-1)T}{A}\Big)\right\rceil - 1. \tag{11}$$

In the plane, each step rotates by $\Delta\theta = 2\pi/l$ and expands radius by b per index; after l steps (one visual turn), the *radial* growth is b^l and the *support* growth in digits follows (**??**). Thus the spiral pitch scales like

$$\text{pitch} \;\propto\; \frac{\Delta\text{angle}}{\Delta \log \text{radius}} \;\sim\; \frac{2\pi}{l \log b} \quad \text{with} \quad \log b \text{ set by } \Lambda \text{ and } r. \tag{12}$$

In particular $R(T) = \Theta(\log_r T)$ gives a logarithmic law for how far the support advances per macrocycle.

8 Fractal spawn and termination (optional)

To model child loops spawned at smaller scales, add levels $\ell = 0, 1, \ldots$ with shrink $s \in (0, 1)$: $z = \sum_{\ell \ge 0} s^\ell \sum_k d_{\ell,k} B^k$. If each level satisfies the decimated constraint and the per–level workload obeys $\sum_\ell s^\ell \sum_t |d_{\ell,t}| q^t < \infty$ for some $q \in (1, b^l)$, then normalization converges per level and the sum converges absolutely. A *macro re–closure* arises when the half–twist condition (**??**) is again met after finitely many levels.

9 Multi–site events: pre–alignment and uniqueness

Let $d^{(0)}$ be the decoded anchor profile for one event, normalized by $\sum_j d_j^{(0)} r^j = 1$. For totals $\{T_s\}$, set $\tilde{d} = \sum_s T_s\, S^{-s} d^{(0)}$ with $(S^{-s}d)_k = d_{k+s}$. Then $\sum_k \tilde{d}_k r^k = \sum_s T_s r^s$ matches (**??**). If the anchor ML decoder is unique at SNR $\ge \gamma$ and the shifts have disjoint (or dominated) overlap within the window at the same SNR, then $\tilde{d}$ is the unique ML solution (hard or soft).

10 Modal extension (coupled tracks)

With weak transverse coupling, diagonalize modes θ to obtain per–mode transfers $M(\theta)$ and decimated recurrences $x_{k+2l}^{(\theta)} - V_l(\theta) x_{k+l}^{(\theta)} + Q^l(\theta) x_k^{(\theta)} = 0$, $V_l(\theta) = \operatorname{tr}(M(\theta)^l)$. The checksum and decoder apply per mode. Perturbations $E(\theta)$ change $V_l(\theta)$ by at most $l\|M\|_2^{l-1}\|E\|_2 + O(\|E\|_2^2)$.

11 Protocols and observables

P0 (calibrate). Measure $P = \mathrm{tr} M$, $Q = \det M$; choose l as supercell length or Floquet denominator. **P1 (Möbius).** Impose APBC (–flux) and two–block macrocycle with (**??**); visualize figure–8 centerline. **P2 (checksum).** Inject T anchor events; decode; verify (**??**) (symmetric) or (**??**) (general). **P3 (mode switch).** Sweep Λ by changing cadence or normalization time; observe circular $\to$ spiral transition. **P4 (pitch law).** Plot $R(T)$ vs. $\log T$ and compare to (**??**); estimate effective b from angular/radial drift. **Platforms.** Digital/FPGA proxy (exact arithmetic); microwave ladder (with –flux); mechanical/acoustic chain.

12 Discussion and limits

We avoid cosmological claims; this is an engineered, falsifiable framework. Topology is a boundary/holonomy choice (APBC vs. PBC); dynamics come from the unit–cell transfer; robustness from a trellis decoder. Clean bifurcation types under Λ depend on nonlinearity and dissipation; the invariant checksum holds after decoding regardless.

Companion reference

A companion paper *Chebyshev–Lucas Error Correction in Decimated Transfer–Matrix Chains* develops the decimation identity, checksum, decoder, uniqueness, and bounds in detail; this note reuses those results.